



Lesson 4.1 — Introduction to Rational Functions

Big idea: A rational function is a polynomial divided by a polynomial. Wherever the bottom is zero, the function does something special — either a vertical asymptote or a hole.

Quick review

Form: $f(x) = P(x) / Q(x)$, where P and Q are polynomials and Q is not the zero polynomial.

Domain: all real x except where $Q(x) = 0$.

Behavior at the zeros of Q : if $(x - c)$ is in Q only, you get a vertical asymptote at $x = c$; if it's also a factor of P (and they cancel completely), you get a hole.

Don't cancel before noting the hole: the hole stays at that x even after cancelation.

Worked example

Worked example. Find domain and any holes/asymptotes for $f(x) = (x^2 - 1) / (x^2 - x - 2)$.

Factor: $(x - 1)(x + 1) / [(x - 2)(x + 1)]$. $(x + 1)$ cancels $\rightarrow$ hole at $x = -1$. $(x - 2)$ remains in denominator $\rightarrow$ vertical asymptote at $x = 2$.

Domain: $x \neq -1$ and $x \neq 2$.

Warm-ups

1. State the domain of $f(x) = 1/(x - 5)$.
2. Is $f(x) = (3x + 2)/x$ a rational function?
3. Where might the rational function $f(x) = (x - 4)/(x^2 - 16)$ have asymptotes or holes?

Practice

1. Find all values where $f(x) = (x^2 - 9) / (x^2 - 5x + 6)$ is undefined, and decide hole or vertical asymptote at each.
2. State the domain of $f(x) = (x + 3) / (x^2 + 4x)$.
3. Sketch (describe) $f(x) = 1/x$ near $x = 0$ and as $x \rightarrow \pm\infty$.

Challenge

1. Build a rational function with vertical asymptotes at $x = 1$ and $x = -2$, a hole at $x = 3$, and y -intercept 1.
2. If $f(x) = p(x)/q(x)$ is a rational function and you cancel a common factor, why does the cancelled factor still produce a hole rather than an asymptote in the resulting graph?



Answer key — Lesson 4.1

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $x \neq 5$, i.e. $(-\infty, 5) \cup (5, \infty)$.
2. Yes — both numerator and denominator are polynomials.
3. Factor: $(x - 4) / [(x - 4)(x + 4)]$. Hole at $x = 4$ (cancels), vertical asymptote at $x = -4$.

Practice

1. Factor: $(x - 3)(x + 3) / [(x - 2)(x - 3)]$. Hole at $x = 3$ (cancels), vertical asymptote at $x = 2$.
2. $x^2 + 4x = x(x + 4) \neq 0 \Rightarrow x \neq 0, x \neq -4$. Domain: $\mathbb{R} \setminus \{0, -4\}$.
3. Near $x = 0$: $f \rightarrow +\infty$ from the right, $-\infty$ from the left. As $x \rightarrow \pm\infty$: $f \rightarrow 0$. Two branches in opposite quadrants.

Challenge

1. $f(x) = (x - 3) / [(x - 1)(x + 2)]$ would have those VAs and a hole at $x = 3$ if $x = 3$ also appears as a factor of the numerator. To get y-intercept 1: try $f(x) = k(x - 3)^2 / [(x - 1)(x + 2)(x - 3)] = k(x - 3) / [(x - 1)(x + 2)]$. $f(0) = k(-3)/[(-1)(2)] = 3k/2 = 1 \Rightarrow k = 2/3$.
2. After cancellation the function has the same value as the simplified form everywhere except at the cancelled x — at that x the original is undefined ($0/0$). So the graph matches the simplified curve except for a single missing point: a hole.